



FACULTY OF COMPUTING

SEMESTER 1/20252026

SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 03

ASSIGNMENT 3

INTELLIGENT AGENT

THEME:

HEALTHCARE

LECTURER:

DR. NORSHAM BINTI IDRIS

Group no: Group 4

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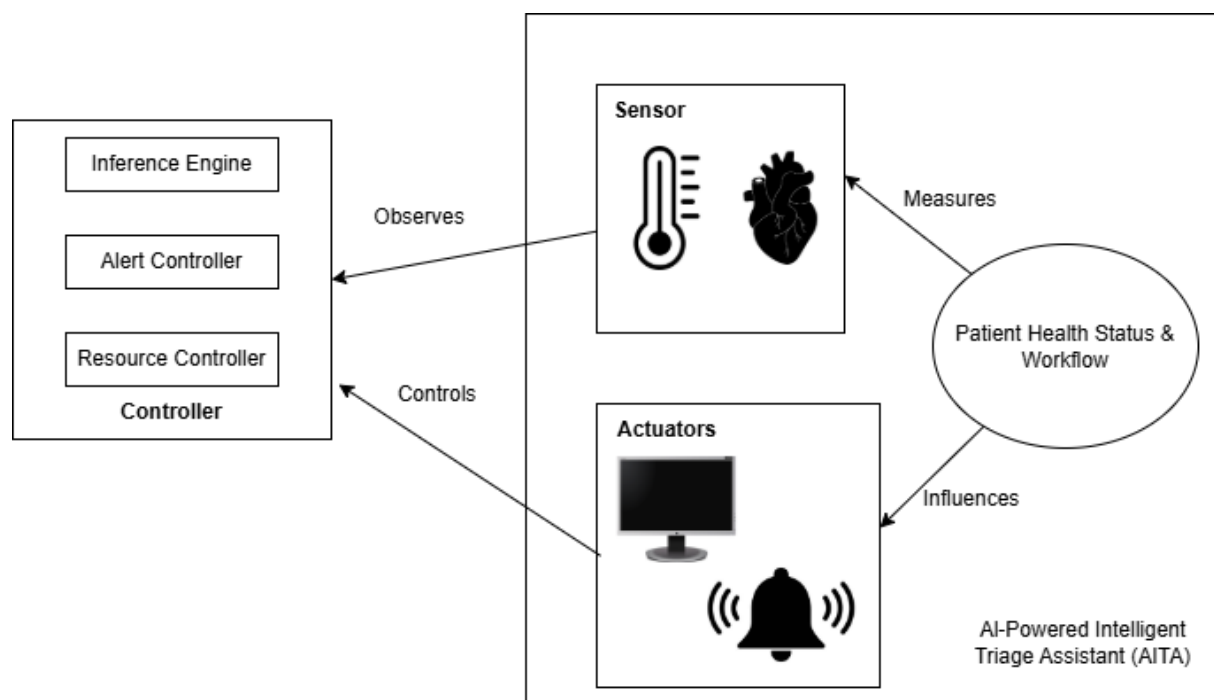
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Intelligent Agent

PEAS Model Representation

Performance	Accuracy of severity scoring, Reduction in patient waiting time, Correct prioritization of critical cases, Successful bed allocation.
Environment	Emergency Department (ED), Triage Station, Hospital Information System (HIS), Patient Waiting Area.
Actuators	Triage Dashboard Display (Severity Score), Visual/Audio Alerts (Shock/Critical warnings), Database Update (Bed Assignment).
Sensors	Digital Thermometer, Pulse Oximeter (SpO2 & Heart Rate), Blood Pressure Monitor, Keyboard/Touchscreen (for manual symptom entry).



Explanations

This system gathers patient data, analyses severity levels, and helps medical personnel make decisions in order to support the triage process in the Emergency Department (ED). The system guarantees that the most urgent cases receive prompt attention, increases patient safety, and streamlines workflow.

Performance

During triage operations, the system's primary goal is to provide accurate and dependable performance. It assesses patient data to produce accurate severity scores, assisting medical personnel in promptly identifying critical cases. The system improves the flow of patients through the ED and lowers overall patient waiting times by automating portions of the assessment process. Additionally, it guarantees the proper prioritization of critical patients. Additionally, by updating the database to match patients with appropriate beds based on their condition and current availability, the system aids in the efficient allocation of hospital beds.

Environment

This system is designed to function in hospital settings, particularly in emergency care-related areas. It is mostly utilized at the triage station, where patient evaluations are conducted, and in the emergency room. Additionally, the system communicates directly with the Hospital Information System (HIS), which maintains bed availability and patient records. Because faster evaluations and better prioritization result in more efficient patient flow and less traffic, its operation also affects the patient waiting area.

Actuators

The system provides outputs through several types of actuators. The triage dashboard displays the patient's severity score, allowing staff to make informed decisions at a glance. Visual and audio alerts are activated when the system detects life-threatening conditions such as shock or extremely unstable vital signs. These alerts ensure that medical personnel can respond immediately. Furthermore, the system updates the hospital database automatically, especially when assigning beds to patients based on the severity level calculated.

Sensors

The system uses a variety of sensors and input devices to collect patient data. These include a blood pressure monitor to evaluate cardiovascular stability, a digital thermometer to measure body temperature, and a pulse oximeter to record heart rate and oxygen saturation. The system uses the vital real-time data from these sensors to assess the patient's condition. Furthermore, a keyboard or touchscreen interface enables medical personnel to manually enter patient complaints, symptoms, and other observations, guaranteeing that the assessment takes into account both automated data and human insight.

Peer review on Team Working (TW)

Category	4	3	2	1
Active participation	Routinely provides useful ideas when participating in the group.	Usually provides useful ideas when participating in the group.	Sometimes provides useful ideas when participating in the group.	Rarely provides useful ideas when participating in the group.

MEMBERS	ELIJAH SHE YU SHENG	LAU YEE WEN	CHERYL CHEONG KAH VOON	MUHAMMAD ADAM BIN RAZALI
ELIJAH SHE YU SHENG	-	4	4	4
LAU YEE WEN	4	-	4	4
CHERYL CHEONG KAH VOON	4	4	-	4
MUHAMMAD ADAM BIN RAZALI	4	4	4	-